Problem set 2: Geometry and gravitation

This problem set contains some exercises that are purely related to differential geometry. These are written using proper notations often disregarded in physics. These are the notations adopted in the notes so, when in doubt, you must refer to those for details. Exercises marked with **(*)** are optional.

1. **Curves, vectors and covectors on the sphere**
   Consider the manifold $S^2 = \{(x, y, z) \in \mathbb{R}^3, x^2 + y^2 + z^2 = 1\}$. Denoting $N = (0, 0, 1) \in S^2$ and $S = (0, 0, -1) \in S^2$, the map:

$$\varphi^{-1} : \left\{ \begin{array}{ccc}]0, \pi[\times]0, 2\pi[& \rightarrow & S^2\backslash\{N, S\} \\ (\theta, \phi) & \mapsto & (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta) \end{array} \right.$$

   defines a local chart $\left\{S^2\backslash\{N, S\}, \varphi\right\}$ on S^2.

   (a) Consider $r \in \mathbb{R}_+^*$ and the curve $c :]0, 2\pi[\rightarrow S^2$ such that:

$$\varphi \circ c : \left\{ \begin{array}{ccc}]0, 2\pi[& \rightarrow & \mathbb{R}^2 \\ t & \mapsto & \left(\frac{t}{r}, t\right). \end{array} \right. . \tag{1-1}$$

   Show that, in $S^2\backslash\{N, S\}$, the tangent vector $\boldsymbol{X_p}$ to the curve c at any point $p \in c(]0, 2\pi[)$ such that $p = \varphi^{-1}(\theta, \phi)$ is given in the local coordinate system, by:

$$\boldsymbol{X_p} = \frac{1}{r}\frac{\partial}{\partial\theta}_{|\varphi(p)} + \frac{\partial}{\partial\phi}_{|\varphi(p)} . \tag{1-2}$$

   (b) Then, we can define a vector field X on $S^2\backslash\{N, S\}$ by

$$\forall p \in S^2\backslash\{N, S\}, \boldsymbol{X}(p) = \boldsymbol{X_p} = \frac{1}{r}\frac{\partial}{\partial\theta}_{|\varphi(p)} + \frac{\partial}{\partial\phi}_{|\varphi(p)} . \tag{1-3}$$

   What curve on S^2 is tangent to $\boldsymbol{X_p}$ at all $p \in S^2\backslash\{N, S\}$?

   (c) Consider $f \in \mathcal{F}(M)$ defined for $p \in S^2\backslash\{N, S\}$ such that $\phi(p) = (\theta, \phi)$ by:

$$f \circ \phi^{-1}(\theta, \phi) = \sin\theta\cos\phi . \tag{1-4}$$

   Show that:

$$\mathrm{d}f_p = \cos\theta\cos\phi\mathrm{d}\theta - \sin\theta\sin\phi\mathrm{d}\phi . \tag{1-5}$$

   (d) Consider the one-form at $p \in S^2\backslash\{N, S\}$, ω_p given, in the local chart, by:

$$\boldsymbol{\omega_p} = -\mathrm{d}\theta + \mathrm{d}\phi . \tag{1-6}$$

   Show that, for the tangent vector defined above:

$$\boldsymbol{\omega_p}(\boldsymbol{X_p}) = 1 - \frac{1}{r^2} . \tag{1-7}$$

2. **Real projective space (*)**
We consider the set of lines through 0 in $\mathbb{R}^3$, called the real projective space of dimension 2, and denoted $\mathbb{R}P^2$.

(a) We introduce a relation $\sim$ in $\mathbb{R}^3$ by:

$$\forall \left((x^0, x^1, x^2), (y^0, y^1, y^n) \right) \in \mathbb{R}^3 \times \mathbb{R}^3, (x^0, x^1, x^2) \sim (y^0, y^1, y^n) \text{ iff } \exists t \in \mathbb{R}, (y^0, y^1, y^n) = t(x^0, x^1, x^2) \ . \tag{2-1}$$

Show that this is an equivalence relation (symmetric, transitive, reflexive relation). Then:

$$\mathbb{R}P^2 = \mathbb{R}^3_* / \sim \ , \tag{2-2}$$

i.e., the elements of $\mathbb{R}P^2$ are the equivalence class of the equivalence relation $\sim$.

(b) We want to show that $\mathbb{R}P^2$ is a manifold of dimension 2. Given a representative of each class of equivalence, say (x^0, x^1, x^2), we define, for $i \in \{0, 1, 2\}$, by U_i the set of lines with $x^i \neq 0$ (there are three of them). Then, we define the inhomogeneous coordinates $\xi_{(i)}$ on U_i, with $p \in U_i$ by:

$$\xi_{(0)}(p) = (\frac{x^1}{x^0}, \frac{x^2}{x^0}) \tag{2-3}$$

$$\xi_{(1)}(p) = (\frac{x^0}{x^1}, \frac{x^2}{x^1}) \tag{2-4}$$

$$\xi_{(2)}(p) = (\frac{x^0}{x^2}, \frac{x^1}{x^2}) \ . \tag{2-5}$$

i. Justify that these coordinates do not depend on the choice of representative of the classes of equivalence.

ii. Show that the three functions $\xi_{(i)}$ define local coordinate functions, and that $\mathbb{R}P^2 = \cup_{i \in \{0,1,2\}} U_i$.

iii. Show that the transition functions $\psi_{ij} = \xi_{(i)} \circ \xi^{-1}_{(j)}$ defined on $U_i \cap U_j$ are smooth. Conclude that $\mathbb{R}P^2$ is indeed a manifold of dimension 2.

(c) Can you generalise this construction to an arbitrary dimension, $\mathbb{R}P^n$?

(d) Consider the sphere $S^2 = \{(x, y, z) \in \mathbb{R}^3, x^2 + y^2 + z^2 = 1\}$. By taking, fo representatives of each element of $\mathbb{R}P^2$ points such that $\left(x^0\right)^2 + \left(x^1\right)^2 + \left(x^2\right)^2 = 1$, show that there is a diffeomorphism between $\mathbb{R}P^2$ and the sphere S^2 on which we identify antipodal points, i.e., between $\mathbb{R}P^2$ and an hemisphere of S^2.

3. **Spin group as a manifold (*)**
Consider the special unitary group of dimension 2:

$$SU(2) = \left\{ U = \begin{pmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{pmatrix}, (\alpha, \beta) \in \mathbb{C}^2, |\alpha|^2 + |\beta|^2 = 1 \right\} \subset M_2(\mathbb{C}) \ . \tag{3-1}$$

(a) Show that $\forall U \in SU(2), UU^\dagger = U^\dagger U = I_2$ (hence the name unitary).

(b) Show that $SU(2)$ is a group for the standard matrix multiplication.

(c) Explain why $SU(2)$ is a manifold of real dimension 4.

(d) Give the details of the 3-sphere S^3 as a subset of $\mathbb{R}^4$. Is it a manifold?

(e) Consider the mapping:

$$f : \left\{ \begin{array}{ccc} \mathbb{C}^2 & \to & M_2(\mathbb{C}) \\ (\alpha, \beta) & \mapsto & \begin{pmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{pmatrix} \end{array} \right. \ . \tag{3-2}$$

Show that $f\left(S^3\right) = SU(2)$ and that f so restricted is a diffeomorphism of S^3 onto $SU(2)$.

4. **Two charts on the sphere**
Show how to go from the spherical coordinate system to the stereographic coordinates from the North pole, where these two charts are defined.

5. **Vector fields in the plane**
Consider the plane E^2 described by the Cartesian chart $(x, y) \in \mathbb{R}^2$. Let:

$$\boldsymbol{X} = x\frac{\partial}{\partial x} - y\frac{\partial}{\partial y} \tag{5-1}$$

$$\boldsymbol{Y} = x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y} \ . \tag{5-2}$$

(a) What are the integral curves tangent to $\boldsymbol{X}$ and $\boldsymbol{Y}$?

(b) Determine the commutator $[\boldsymbol{X}, \boldsymbol{Y}] = XY - YX$.

6. **Conformal transformations**
Given a manifold $\mathcal{M}$ equipped with a metric $\boldsymbol{g}$, a conformal transformation of the metric tensor is defined by the mapping:

$$\boldsymbol{g}(p) \mapsto f(p)\boldsymbol{g}(p) \ , \tag{6-1}$$

for $p \in \mathcal{M}$ and $f \in \mathcal{F}(\mathcal{M})$ an arbitrary function.

Show that a conformal transformation preserves the angle between any two vectors.

(a) Show that lightlike curves remain lightlike under a conformal transformation. Is it true for timelike curves?

7. **Local Lorentz factor and observables**
Follow the developments in sections 3.5.4 and 3.5.5 of the lecture notes.

8. **Affine connection**
Here are a few important exercises from the notes on affine connections.

(a) Following section 3.6.1 of the lecture notes, derive Eqs (3.126)-(3.127) giving the connection coefficients for the standard parallel transport in $\mathbb{R}^2$ in polar coordinates.

(b) Determine how connection coefficients change under a change of coordinates, Eq. (3.152) of the lecture notes. Conclude that connection coefficients are not the components of a tensor field.

(c) Follow the calculations presented on pages 133 and 134 of the notes, which show that timelike geodesics between two events p and q are curves that extremise the proper time elapsed along the curves.

(d) Reproduce the results of section 3.6.4 of the notes on geodesics in a static, weak field.

(e) Follow the developments in section 3.7.1 of the notes establishing the geodesic deviation equation and the expression of the Riemann curvature tensor.

(f) Reproduce the results of section 3.7.3 of the notes which provide a clear interpretation of Riemann curvature in terms of tidal distortions.

9. **Index manipulations etc.**

   (a) For a general metric $g_{\alpha\beta}$ whose determinant is g, show that:

   $$\partial_\alpha g = g g^{\beta\gamma} \partial_\alpha g_{\beta\gamma} \ . \tag{9-1}$$

   (b) Deduce the following:

   1. $\Gamma^{\beta}_{\beta\alpha} = \frac{1}{2}\partial_\alpha(\ln|g|)$,
   2. $g^{\beta\gamma}\Gamma^{\alpha}_{\beta\gamma} = -[\partial_\beta(g^{\alpha\beta}\sqrt{-g})]/\sqrt{-g}$,
   3. $\nabla_\alpha v^\alpha = [\partial_\alpha(\sqrt{-g}v^\alpha)]/\sqrt{-g}$
   4. for an antisymmetric tensor $F_{\alpha\beta}$, $\nabla_\beta F^{\alpha\beta} = [\partial_\beta(\sqrt{-g}F^{\alpha\beta})]/\sqrt{-g}$.

10. **Parallel transport on the sphere**
    We consider the manifold $S^2 = \{(x, y, z) \in \mathbb{R}^3, x^2 + y^2 + z^2 = 1\}$.

    (a) By introducing spherical coordinates (θ, ϕ) on the sphere as previously, show that the Euclidean metric on $\mathbb{R}^3$, given in Cartesian coordinates, by the line element:

    $$\mathrm{d}s^2 = \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 \ , \tag{10-1}$$

    induces the *round metric* on S^2, given by:

    $$\mathrm{d}s^2 = \mathrm{d}\theta^2 + \sin^2\theta \mathrm{d}\phi \ . \tag{10-2}$$

    (b) Determine the connection coefficient for the metric-compatible, symmetric connection:

    $$\Gamma^{\theta}{}_{\phi\phi} = -\frac{1}{2}\sin 2\theta \ ; \ \Gamma^{\phi}_{\phi\theta} = \frac{\cos\theta}{\sin\theta} \ . \tag{10-3}$$

    (c) Write down the geodesic equations and find their general solution.

    (d) Compute the Riemann tensor.

    (e) A vector $\boldsymbol{V}$ is parallel transported along a geodesic from a point N on the equator to the North pole, then from the North pole to another point B of the equator. Show that during that transport, $\boldsymbol{V}$ is rotated by an angle α which is proportional to the area of the triangle (on the sphere, i.e. geodesic) ANB.

    (f) Determine the connection coefficients in the orthonormal, non-coordinate basis made of $\boldsymbol{e}_{(1)} = \frac{\partial}{\partial\theta}$ and $\boldsymbol{e}_{(2)} = \frac{1}{\sin\theta}\frac{\partial}{\partial\phi}$.

    (g) Show that the parallel transport of a vector $\boldsymbol{U}$ along a loop γ on the sphere (not necessarily made of bits of geodesics) results in a rotation of the vector by an angle ω equal to the area of the interior of the loop.

11. **The cylinder is flat**
    Draw a picture of a cylinder. On it, draw a closed curve which goes once around the cylinder; draw another which doesn't. Parallel transport a vector around these loops (in pictures!). Assuming that curvature is related to how much a vector rotates when transported around a closed loop, deduce that a cylinder is in this sense flat. Why? (How do you make a cylinder?) Compare with the case of a sphere.

12. **2D de Sitter spacetime**

We consider the manifold:

$$M = \left\{(x, y, v) \in \mathbb{R}^3, -v^2 + x^2 + y^2 = \alpha^2\right\} \ , \tag{12-1}$$

for $\alpha \in \mathbb{R}_+^*$.

(a) Show that the map:

$$\phi : \left\{ \begin{array}{ccc} M\backslash\{\left(\sqrt{\alpha^2+v^2}, 0, v\right), v \in \mathbb{R}\} & \to &]0, 2\pi[\times\mathbb{R} \\ (x, y, v) & \mapsto & (\theta, t) = \left(\arccos\left(\frac{x}{\sqrt{\alpha^2+v^2}}\right), \alpha \operatorname{arcsinh}\left(\frac{v}{\alpha}\right)\right) \end{array} \right. \tag{12-2}$$

defines a local chart on $M\backslash\{\left(\sqrt{\alpha^2+v^2}, 0, v\right), v \in \mathbb{R}\}$, with its inverse given by:

$$\phi^{-1} : \left\{ \begin{array}{ccc}]0, 2\pi[\times\mathbb{R} & \to & M\backslash\{\left(\sqrt{\alpha^2+v^2}, 0, v\right), v \in \mathbb{R}\} \\ (\theta, t) & \mapsto & \left(\alpha \cosh\left(\frac{t}{\alpha}\right)\cos\theta, \alpha\cosh\left(\frac{t}{\alpha}\right)\sin\theta, \alpha\sinh\left(\frac{t}{\alpha}\right)\right) . \end{array} \right. \tag{12-3}$$

(b) Show that the standard Minkowskian metric of $\mathbb{R}^3$:

$$\boldsymbol{G} = \mathrm{d}x \otimes \mathrm{d}x + \mathrm{d}y \otimes \mathrm{d}y - \mathrm{d}v \otimes \mathrm{d}v \tag{12-4}$$

induces, in the coordinate chart (θ, t) the following metric on M:

$$\boldsymbol{g} = -\mathrm{d}t \otimes \mathrm{d}t + \alpha^2 \cosh^2\left(\frac{t}{\alpha}\right) \mathrm{d}\theta \otimes \mathrm{d}\theta \ . \tag{12-5}$$

(c) Show that, if one introduces a new coordinate η such that:

$$\eta = 2\arctan\left(\mathrm{e}^{\frac{t}{\alpha}}\right) - \frac{\pi}{2} \ , \tag{12-6}$$

with $\eta \in]-\pi/2, \pi/2[$, the coordinate system (η, θ) also covers $M\backslash\{\left(\sqrt{\alpha^2+v^2}, 0, v\right), v \in \mathbb{R}\}$ and the metric can then be written:

$$\boldsymbol{g} = F(\eta)\left(-\mathrm{d}\eta \otimes \mathrm{d}\eta + \mathrm{d}\theta \otimes \mathrm{d}\theta\right) \ , \tag{12-7}$$

for a function F to be determined. This shows that the de Sitter spacetime is conformally flat.
Hint: Recall that $\frac{d}{dt}\arctan(f(t)) = f'(t)/(1+f^2(t))$.

(d) Show that, in the (θ, t) coordinates, the only non-zero components of the Levi-Civita connection associated to g are given by:

$$\Gamma^t{}_{\theta\theta} = \alpha \sinh\left(\frac{t}{\alpha}\right)\cosh\left(\frac{t}{\alpha}\right) \tag{12-8}$$

$$\Gamma^\theta{}_{t\theta} = \Gamma^\theta{}_{\theta t} = \frac{\sinh\left(\frac{t}{\alpha}\right)}{\alpha\cosh\left(\frac{t}{\alpha}\right)} \ . \tag{12-9}$$

(e) Deduce that the geodesic equations read:

$$\frac{\mathrm{d}^2 t}{\mathrm{d}\lambda^2} + \alpha\sinh\left(\frac{t}{\alpha}\right)\cosh\left(\frac{t}{\alpha}\right)\left(\frac{\mathrm{d}\theta}{\mathrm{d}\lambda}\right)^2 = 0 \tag{12-10}$$

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}\lambda^2} + \frac{2}{\alpha}\frac{\sinh\left(\frac{t}{\alpha}\right)}{\cosh\left(\frac{t}{\alpha}\right)}\frac{\mathrm{d}t}{\mathrm{d}\lambda}\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = 0, \tag{12-11}$$

where λ is a parameter along the geodesics c, such that the tangent vector to the geodesics reads: $\boldsymbol{X} = \frac{\mathrm{d}x^i(c(\lambda))}{\mathrm{d}\lambda}\boldsymbol{e}_{(i)}$.

(f) By integrating Eq. (12-11), show that there exists a constant $J \in \mathbb{R}$ such that:

$$\frac{d\theta}{d\lambda} = \frac{J}{\alpha^2 \cosh^2\left(\frac{t}{\alpha}\right)}. \tag{12-12}$$

(g) By using the previous result, and integrating Eq. (12-10) once with respect to λ, show that, there exists a constant $E \in \mathbb{R}$ such that:

$$E = \frac{1}{2}\left(\frac{\mathrm{d}t}{\mathrm{d}\lambda}\right)^2 - \frac{1}{2}\left(\frac{J}{\alpha}\right)^2 \frac{1}{\cosh^2\left(\frac{t}{\alpha}\right)} \ . \tag{12-13}$$

(h) Deduce that E is bounded from below:

$$E \geq -\frac{1}{2}\left(\frac{J}{\alpha}\right)^2 \ . \tag{12-14}$$

(i) By calculating $\boldsymbol{g}(\boldsymbol{X}, \boldsymbol{X})$ where $\boldsymbol{X}$ is the tangent vector field to the geodesics, determine under which condition the geodesics is timelike, spacelike or null.

(j) Find the solution to the geodesics equations for $J = 0$. What kind of geodesics is it?

(k) Find the solution to the geodesics equations for $E = 0$.
Hint: Use that $\frac{\mathrm{d}}{\mathrm{d}x}\arctan(x) = \frac{1}{1+x^2}$.

13. **Lie derivative: an exercise**
For any two vector fields, $\boldsymbol{u}$ and $\boldsymbol{v}$, their *Lie bracket* is defined to be the vector field $[\boldsymbol{u}, \boldsymbol{v}]$ with components (this is NOT the proper definition):

$$[\boldsymbol{u}, \boldsymbol{v}]^\alpha = u^\beta \nabla_\beta v^\alpha - v^\beta \nabla_\beta u^\alpha \ . \tag{13-1}$$

Show that $[\boldsymbol{u}, \boldsymbol{v}] = -[\boldsymbol{v}, \boldsymbol{u}]$. Show that we can calculate the Lie bracket without any reference to the metric whatsoever.

The Lie bracket is related to the Lie derivative of $\boldsymbol{v}$ along $\boldsymbol{u}$, which is defined by:

$$\mathcal{L}_{\boldsymbol{u}} \boldsymbol{v} = [\boldsymbol{u}, \boldsymbol{v}], \quad \mathcal{L}_{\boldsymbol{u}} f = \boldsymbol{\nabla}_{\boldsymbol{u}} f \ , \tag{13-2}$$

where f is any scalar, Show that it is a derivative operator by showing that $\mathcal{L}_{\boldsymbol{u}}(f\boldsymbol{v})$ obeys the Leibniz rule for any scalar f. The Lie derivative is the derivative of $\boldsymbol{v}$ along $\boldsymbol{u}$, and can be defined without reference to a metric: it therefore plays a fundamental role in differential geometry.

14. **The Einstein-Hilbert action**
The Einstein field equations can be recovered using an action principle, if one starts from the *Einstein-Hilbert action*:

$$S_{EH} = \frac{1}{16\pi G} \int R\,[g^{\mu\nu}]\,\sqrt{-g}\mathrm{d}x^4 + S_M \ , \tag{14-1}$$

where $g = \det \boldsymbol{g}$ and $S_M = \int \mathcal{L}_M \sqrt{-g}\mathrm{d}^4x$ is the action functional for the matter content. In order to achieve this, we perturb the Einstein-Hilbert action around a given metric by writing $g_{\mu\nu} = \bar{g}_{\mu\nu} + \delta g^{\mu\nu}$.

(a) Using Jacobi's formula for invertible matrices:

$$\det\,(A + \varepsilon B) = \det A\,\left[1 + \varepsilon \mathrm{Tr}(A^{-1}B)\right] + O(\varepsilon) \ , \tag{14-2}$$

show that:

$$\sqrt{-g} = \sqrt{-\bar{g}} - \frac{1}{2} g_{\alpha\beta}\delta g^{\alpha\beta}\sqrt{-\bar{g}} + O(\boldsymbol{\delta g}) \ . \tag{14-3}$$

(b) Writing

$$\Gamma^{\mu}{}_{\nu\rho} = \bar{\Gamma}^{\mu}{}_{\nu\rho} + \delta\Gamma^{\mu}{}_{\nu\rho} \ , \tag{14-4}$$

justify that $\delta\Gamma^{\mu}{}_{\nu\rho}$ are the components of a tensor.

(c) Deduce that the variation in the Riemann tensor is:

$$\delta R^{\rho}{}_{\sigma\mu\nu} = \nabla_{\mu}\delta\Gamma^{\rho}{}_{\nu\sigma} - \nabla_{\nu}\delta\Gamma^{\rho}{}_{\mu\sigma} \ . \tag{14-5}$$

(d) Thus, show that:

$$\delta R = \bar{R}_{\sigma\nu}\delta g^{\sigma\nu} + \nabla_{\rho}\left[g^{\sigma\nu}\delta\Gamma^{\rho}{}_{\nu\sigma} - g^{\sigma\rho}\delta\Gamma^{\nu}{}_{\nu\sigma}\right] \ . \tag{14-6}$$

(e) Now, show that:

$$\Gamma^{\alpha}{}_{\mu\alpha} = \frac{1}{\sqrt{-g}}\frac{\partial\sqrt{-g}}{\partial x^{\mu}} \ , \tag{14-7}$$

so that for any vector $\boldsymbol{A}$:

$$\sqrt{-g}\nabla_{\mu}\left[A^{\mu}\right] = \frac{\partial}{\partial x^{\mu}}\left[\sqrt{-g}A^{\mu}\right] \ . \tag{14-8}$$

(f) Deduce that, up to a boundary term that we will set to zero (how and why?), we get:

$$\delta S_{EH} = \frac{1}{16\pi G}\int\left[R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}\right]\delta g^{\mu\nu}\sqrt{-g}\mathrm{d}^4x + \int\frac{\delta\left[\mathcal{L}_M\sqrt{-g}\right]}{\delta g^{\mu\nu}}\delta g^{\mu\nu}\mathrm{d}^4x \ . \tag{14-9}$$

(g) Deduce the Einstein field equations for the energy-momentum tensor:

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta\left[\mathcal{L}_M\sqrt{-g}\right]}{\delta g^{\mu\nu}} \ . \tag{14-10}$$

How would you include the cosmological constant?

15. **Oppenheimer-Volkov model (*)**

We consider a relativistic, static and spherically symmetric star whose interior is supposed to be a perfect fluid characterised by its energy density ρ and pressure p. We assume that the metric inside the star can be written:

$$\mathrm{d}s^2 = g_{\mu\nu}\mathrm{d}x^{\mu}\mathrm{d}x^{\nu} = -\mathrm{e}^{2\Phi(r)}\mathrm{d}t^2 + \mathrm{e}^{2\lambda(r)}\mathrm{d}r^2 + r^2\left[\mathrm{d}\theta^2 + \sin^2\theta\mathrm{d}\phi^2\right] \tag{15-1}$$

for some functions $\Phi(r)$, $\lambda(r)$. Next we suppose by symmetry that $\rho(r)$ and $p(r)$ and we write the energy-momentum tensor of matter inside the star as:

$$T_{\mu\nu} = (\rho + p)\,u_{\mu}u_{\nu} + pg_{\mu\nu} \ , \tag{15-2}$$

where $u^{\mu}(r)$ are the components of the 4-velocity of the fluid.

(a) Show that $u^{\mu} = \mathrm{e}^{-\Phi}\delta^{\mu}{}_{0}$.

(b) Write the components of the energy-momentum tensor explicitly.

(c) Show that the conservation fo energy-momentum reduces to:

$$(\rho + p)\,\frac{\mathrm{d}\Phi}{\mathrm{d}r} = -\frac{\mathrm{d}p}{\mathrm{d}r} \ , \tag{15-3}$$

which expresses hydrostatic equilibrium.

(d) Show that the Einstein field equations read:

$$\begin{cases} \left(2r\dfrac{\mathrm{d}\lambda}{\mathrm{d}r} - 1\right)\mathrm{e}^{-2\lambda} + 1 = 8\pi G r^2 \rho & (15\text{-}4) \\ \left(2r\dfrac{\mathrm{d}\Phi}{\mathrm{d}r} + 1\right)\mathrm{e}^{-2\Lambda} - 1 = 8\pi G r^2 p & (15\text{-}5) \\ \dfrac{\mathrm{d}^2\Phi}{\mathrm{d}r^2} + \left(\dfrac{\mathrm{d}\Phi}{\mathrm{d}r}\right)^2 - \dfrac{\mathrm{d}\Phi}{\mathrm{d}r}\dfrac{\mathrm{d}\lambda}{\mathrm{d}r} + \dfrac{1}{r}\left(\dfrac{\mathrm{d}\phi}{\mathrm{d}r} - \dfrac{\mathrm{d}\Lambda}{\mathrm{d}r}\right) = 8\pi G p \mathrm{e}^{2\lambda} \,. & (15\text{-}6) \end{cases}$$

(e) Let us suppose that the star has (areal) radius R such that, for $r > R$, $\rho(r) = p(r) = 0$. What should be the geometry outside the star?

(f) By analogy with the Schwarzschild metric, let us write, inside the star, for $r \leq R$:

$$\mathrm{e}^{-2\lambda} = 1 - \frac{2Gm(r)}{r} \,, \tag{15-7}$$

for a "mass" function $m(r)$. Show that Eq. (15-4) becomes:

$$\frac{\mathrm{d}m}{\mathrm{d}r} = 4\pi r^2 \rho \,. \tag{15-8}$$

Deduce an interpretation of the function $m(r)$.

(g) Show that Eqs. (15-5) then read:

$$\frac{\mathrm{d}\Phi}{\mathrm{d}r} = \left(1 - \frac{2Gm(r)}{r}\right)^{-1}\left(\frac{Gm(r)}{r^2} + 4\pi G r p(r)\right) \,. \tag{15-9}$$

Do we need Eq. (15-6)? Why? The system Eqs (15-3)-(15-8)-(15-9) is known as the *Tolman-Oppenheimer-Volkov system*. In order to solve it. one needs to prescribe an equation of state for the star, $p = p(\rho)$.

(h) We will now solve this system for the particular (unphysical) case of an incompressible star with:

$$\rho(r) = \rho_0 = \text{constant.} \tag{15-10}$$

i. Determine $m(r)$ in terms of R and of the total mass of the star $M = m(R)$. Why is it necessary to fix $m(0) = 0$?

ii. Show that Eq. (15-3) becomes:

$$(\rho_0 + p(r))\,\mathrm{e}^{\Phi(r)} = \rho_0\sqrt{1 - \frac{R_S}{R}} \,, \tag{15-11}$$

where v$R_S = 2GM$.

iii. Show that:

$$\frac{1}{r}\frac{\mathrm{d}\Phi}{\mathrm{d}r} = \frac{R_S}{R^3}\left(1 - \frac{R_S r^2}{R^3}\right)^{-1}\left[\frac{3}{2}\mathrm{e}^{-\Phi}\sqrt{1 - \frac{R_S}{R}} - 1\right] \,. \tag{15-12}$$

iv. Show that the variables:

$$x = \sqrt{1 - \frac{R_S r^2}{R^3}} \text{ and } N = \mathrm{e}^{\Phi} \tag{15-13}$$

obey:

$$x\frac{\mathrm{d}N}{\mathrm{d}x} - N = \frac{3}{2}\sqrt{1 - \frac{R_S}{R}} \,. \tag{15-14}$$

Solve this equation and show that for $0 \leq r \leq R$:

$$\mathrm{e}^{-\Phi(r)} = \frac{3}{2}\sqrt{1 - \frac{R_S}{R}} - \frac{1}{2}\sqrt{1 - \frac{R_S r^2}{R^2}} \,. \tag{15-15}$$

v. Deduce the distribution of pressure $p(r)$ inside the star. What is $p(R)$? And $p_c = p(0)$? Give an expression for p_c in terms of ρ_0 and the *compactness* of the star:

$$C = \frac{R_S}{2R} \, . \tag{15-16}$$

Plot p_c as a function of C at ρ_0 fixed. What happens for $C = 4/9$? Deduce that such a star cannot have a ratio R_S/R larger than a critical value that you will determine. What is the maximum mass of such a star? Give a numerical estimate of this maximum mass for $\rho_0 = 2 \times 10^{17}\,\mathrm{kg\,m^{-3}}$ (nuclear density).

vi. What is the limit of $\mathrm{e}^{2\Phi(0)}$ as $C \to 4/9$?

vii. Determine the redshift experienced by a photon emitted from the surface of the star and received by a static observer at infinity. What is the maximum value of this redshift?

16. **Rindler spacetime**

We consider the 2-dimensional Rindler spacetime given in terms of coordinates (t, x) with $t \in \mathbb{R}$ and $x \in]0, +\infty[$, by the line element:

$$\mathrm{d}s^2 = -x^2 \mathrm{d}t^2 + \mathrm{d}x^2 \, . \tag{16-1}$$

(a) Why is x confined to be strict positive?

(b) Show that null geodesics in this spacetime obey:

$$\left(\frac{\mathrm{d}t}{\mathrm{d}x}\right)^2 = \frac{1}{x^2} \, , \tag{16-2}$$

and solve this equation.

(c) Define adapted null coordinates (u, v) such that the metric becomes:

$$\mathrm{d}s^2 = -e^{v-u} \mathrm{d}u \mathrm{d}v \, . \tag{16-3}$$

What is the possible range of these coordinates?

(d) In order to try and extend our spacetime past the 'singularity' $x = 0$, define a new coordinate system (U, V) such that:

$$\mathrm{d}s^2 = -\mathrm{d}U \mathrm{d}V \, . \tag{16-4}$$

What are the possible ranges of U and V?

(e) Show that the spacetime covered by (U, V) is actually Minkowski spacetime. Draw the spacetime diagram and represent the lines of constant x and t in this diagram.

(f) Show that Rindler spacetime is actually Minkowski spacetime seen by an observer experiencing a constant acceleration with respect to the standard inertial frame.

17. **Warp-drive spacetime**.

Consider a spacetime with coordinates (t, x, y, z). Let $x_s(t)$ be a curve in that spacetime, passing through the origin of the coordinate system and such that the line element reads:

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + [\mathrm{d}x - V_s(t) f(r_s) \mathrm{d}t]^2 + \mathrm{d}y^2 + \mathrm{d}z^2 \, , \tag{17-1}$$

where $V_s(t) = \frac{\mathrm{d}x_s}{\mathrm{d}t}$ is the velocity associated with the curve and $r_s = \sqrt{(x - x_s(t))^2 + y^2 + z^2}$. f is a smooth function with $f(0) = 1$ and that decreases away from the origin to vanish for $r_s > R$ for some R (compact support in technical term).

(a) What is the geometry of slices of spacetime at constant t? Give an interpretation of r_s.

(b) Draw a 2D spacetime diagram by projecting a small, local, portion of the spacetime on your sheet of paper.

(c) Determine the local lightcones and represent the effect of f on the spacetime diagram drawn previously.

(d) Why is it called a warp-drive?

(e) Can this warp-drive be supported by standard matter?